

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459675.7601 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.068 +/- 0.048
Transit depth (Rp/Rs)^2	0.47 +/- 0.66 [%]
Orbital Inclination (inc)	85.89 +/- 1.75
Airmass coefficient 1 (a1)	1.295 +/- 0.017
Airmass coefficient 2 (a2)	-0.253 +/- 0.065
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	None
Optimal Aperture	3.87
Optimal Annulus	12.71
Transit Duration (day)	0.029 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.068 ± 0.048 vs 0.1406 (deviation 1.51σ)
Duration fit vs published	0.029 d vs 0.0975 d (deviation 3.42σ)
Mid-time O–C	1.8 min
Depth SNR	1.4
Residual scatter	1.33 %

Light curve

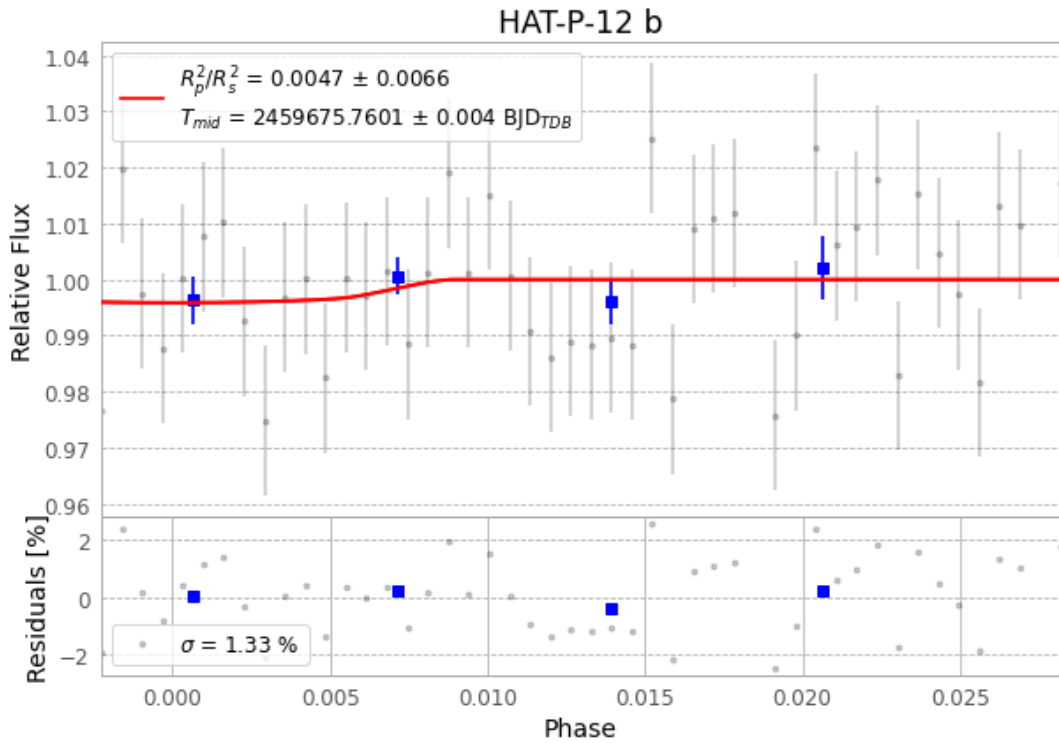


Figure 1 — FinalLightCurve_HAT-P-12 b_2022-04-05.png (SHA-256 56541304d8f7f7cb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [182, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3396a4c115814ae5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

48 FITS frames (32 MB), HATP-12220406055712.FITS → HATP-12220406081814.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-220406.log (7b2de7ab6d9f...), FinalLightCurve_HAT-P-12 b_2022-04-05.png (56541304d8f7...), FinalLightCurve_HAT-P-12 b_2022-04-05.csv (dc301a6f6739...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 22:49Z.